# Orientation and report writing

Fall 2026 · Prof. Ehsan Roohi
September 9, 2026

MEASURE
MODEL
COMPARE

# Experimental evidence and engineering judgment

**01**

## OBSERVE →

See the physical behavior worth explaining

**02**

## MEASURE →

Acquire data with a deliberate measurement system

**03**

## COMPARE →

Test data against theory and models

**04**

## COMMUNICATE

Write a report another engineer can trust

**The goal is not merely to collect numbers. The goal is to make a claim that the data support.**

# Today’s 50-minute session

| | |
|---|---|
| 01 | **0–8 min: course organization and workflow** |
| 02 | 8–20 min: six experiments and their evidence |
| 03 | 20–42 min: report writing and comparison |
| 04 | 42–48 min: policies and report exercise |
| 05 | 48–50 min: first dates and preparation |

# People, places, and weekly rhythm

## COURSE TEAM

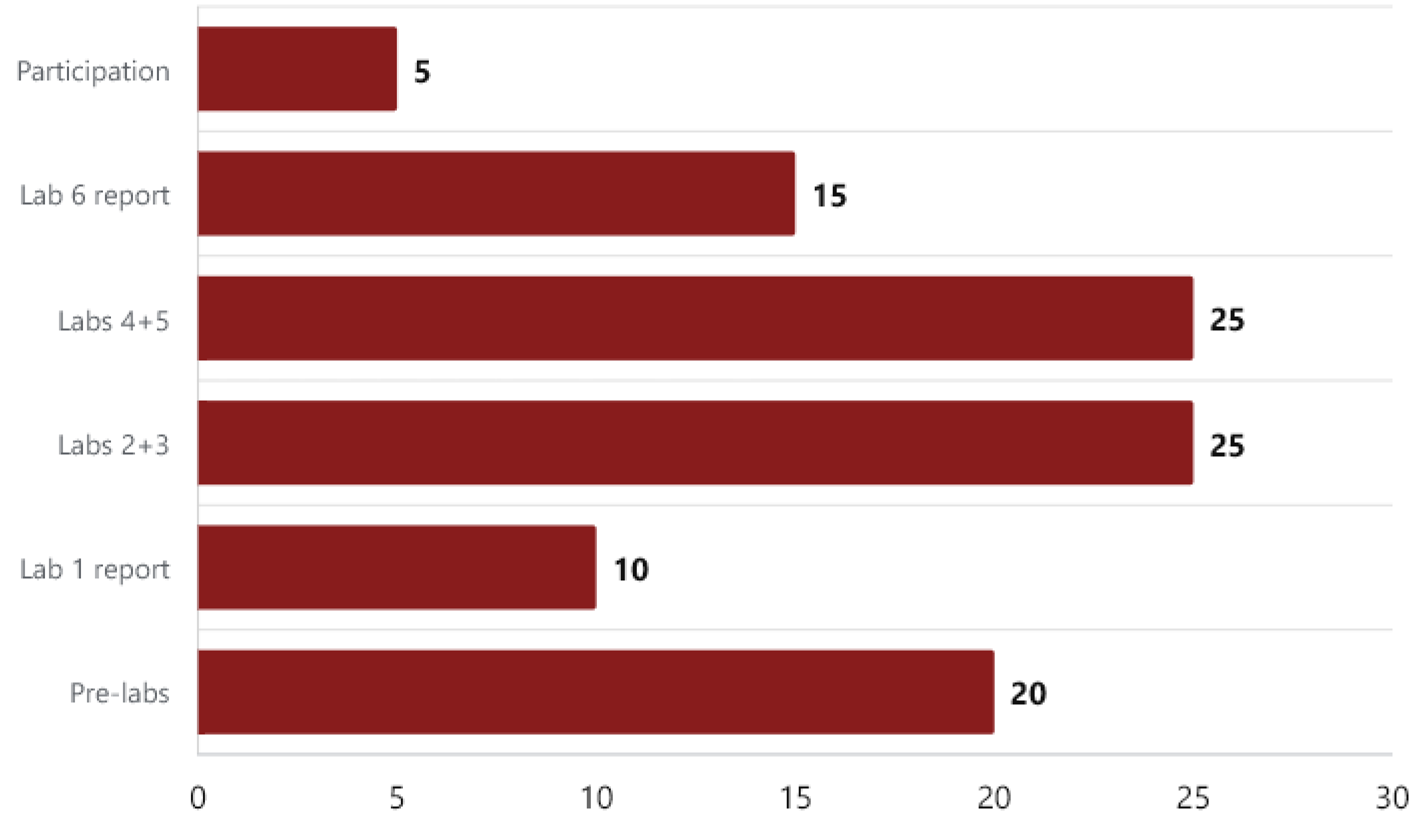
Prof. Ehsan Roohi
Yuyuan Zhang
Ryan Cameron
Nazila Emamdoost

## WEEKLY RHYTHM

| | |
|---|---|
| **MON** | Theory • 12:20–1:10 PM<br>Hasbrouck Laboratory 134 |
| **TUE–FRI** | Laboratory sections<br>Gunness Laboratory 5 |
| **FRI** | Office hours • 12:20–1:10 PM<br>Gunness Laboratory 1 |

**Check Canvas for your assigned section, GTA, and any section-specific updates.**

# Course grading




**100%**

**Preparation + reports + participation**

The combined reports reward synthesis across related experiments.

# The laboratory workflow

**PRE-LAB**

Build the theory and analysis tools before arriving

→

**LAB**

Acquire data, check quality, and document the method

→

**ANALYSIS**

Process data, quantify uncertainty, compare cases

→

**REPORT**

Make a clear claim and support it with evidence

**Pre-labs are due before your own laboratory section begins.**

# Six experiments and four reports

| Lab | Experiment | Primary focus | Report |
|---|---|---|---|
| 1 | Dynamic data sampling | Sampling, aliasing, spectra | Lab 1 |
| 2 | Single pendulum | Dynamic response, model–data | Labs 2+3 |
| 3 | Double pendulum | Coupled and nonlinear motion | Labs 2+3 |
| 4 | 1-DOF free response | Natural frequency, damping | Labs 4+5 |
| 5 | 1-DOF forced response | Resonance, frequency response | Labs 4+5 |
| 6 | Radiometer | Movement measurement | Lab 6 |

# Lab 1: Sampling and frequency analysis

## Question

How can the same sound produce different sampled signals?

## Measurement

Record microphone voltage and the sampling rate. Compare time traces and spectra.

## Report evidence

Show how sample rate and record duration affect the inferred frequency. Identify aliasing.

**$f_s > 2\,fmax$ $\quad\Delta f = f_s / N = 1 / Trecord$**

The Nyquist condition assumes a band-limited input. Practical acquisition needs filtering and margin.

# Lab 2: Single pendulum

## Question

When does a small-angle model predict the measured motion?

## Measurement

Track angle from video. Record geometry, mass properties, frame rate, and release angle.

## Report evidence

Compare measured and predicted angle histories and periods for small and larger releases.

$$I_p \ddot{\theta} + m g d \sin\theta = 0 \qquad T_{small} = 2\pi \sqrt{I_p / m g d}$$

$I_p$ is inertia about the pivot. d is the pivot-to-center-of-mass distance. The model neglects losses.

# Lab 3: Double pendulum

## Question

How does coupling change the motion of two connected bodies?

## Measurement

Track both angles and record both initial angles and release conditions.

## Report evidence

Overlay measured and simulated angle histories. Compare small and large motions and their spectra.

Trajectory divergence alone does not establish chaos. Check initial conditions, tracking, and model assumptions.

# Lab 4: Free response of a 1-DOF system

### Question

What do oscillation period and amplitude decay reveal about the system?

### Measurement

Displace the cart and release it. Measure mass, effective stiffness, and displacement versus time.

### Report evidence

Estimate natural frequency and damping. Compare cases with changed mass, springs, or damping.

$$m\ddot{x} + c\dot{x} + kx = 0 \qquad \omega_n = \sqrt{k / m} \qquad \zeta = c / (2\sqrt{k m})$$

For an underdamped viscous model, $\omega_d = \omega_n \sqrt{1 - \zeta^2}$. Natural and damped frequency differ.

# Lab 5: Forced response of a 1-DOF system

## Question

How does excitation frequency affect response amplitude and phase?

## Measurement

Apply the assigned impulse and harmonic inputs. Record input motion or force and the cart response.

## Report evidence

Separate the transient from the periodic response. Plot amplitude or gain against excitation frequency.

$$m\ddot{x} + c\dot{x} + kx = F(t)$$

This is the force-input model. A moving spring attachment requires the corresponding base-motion input model.

# Lab 6: Radiometer rotation

## Question

How does illumination relate to measured rotation over a controlled time window?

## Measurement

Track cumulative angle from video. Record illuminance, light position, and thermal history.

## Report evidence

Fit angle versus time to estimate RPM. Compare identical measurement windows and discuss warming.

$$\theta(t) \approx \theta_0 + \omega avg\ t \qquad RPM = (60 / 2\pi)\ \omega avg$$

$\omega avg$ is in rad/s. Lux measures illuminance, not absorbed heating power.

# The structure of a laboratory report

### REQUIRED SECTIONS

Abstract
1. Introduction
2. Method
3. Results
4. Comparison
5. Discussion
6. Conclusions

References • Appendices

### WHAT THE READER SHOULD FIND

- A clear purpose and relevant theory
- A reproducible method
- Readable figures with units and labels
- Direct comparison with theory or a model
- A discussion of discrepancies and limitations
- Conclusions supported by the presented data

**Meaningful results matter more than a polished page with weak analysis.**

# Report length and section purpose

| Section | Template allocation | Main purpose |
|---|---|---|
| Abstract | Maximum 10 lines | Purpose, method, key finding, implication |
| Introduction | 1 page | Theory, experiment, and testable objective |
| Method | 1 page | Enough detail to reproduce the measurement |
| Results | 3 pages including figures | Observed behavior and processed measurements |
| Comparison | 1 page | Prediction and experiment on the same basis |
| Discussion | 1 page | Discrepancies, uncertainty, and limitations |
| Conclusions | 1 page | Quantitative answer and the main lesson |

References and optional appendices have no stated page limit. Follow any assignment-specific limit on Canvas.

# Introduction and reproducible method

## Introduction

State a testable objective. Give the governing equation, define symbols, and explain assumptions.

## Method

Describe the actual apparatus, calibration, sampling settings, initial conditions, and repeated runs.

## Analysis record

Explain processing choices, fitting intervals, and parameter sources so another person can repeat the analysis.

Example objective: test how release angle changes agreement with the small-angle pendulum model.

# A concise, quantitative abstract

### Purpose and method

We tested a small-angle pendulum prediction using angle histories from video.

### Key result

The measured frequency was 1.47 Hz, compared with a predicted 1.50 Hz, a 2.0% difference.

### Meaning and limit

The agreement supports the model for this case. The discrepancy requires an uncertainty assessment.

Illustrative wording and invented teaching values. An actual abstract must use your own results. Maximum 10 lines.

# Results: the measured behavior

### Observation

Describe what the measurements show, with quantities and units.

### Evidence

Use figures or tables that answer the experimental question. Identify conditions and repeated runs.

### Traceability

Distinguish measured data, processed measurements, and simulation predictions.

Example observation: successive positive displacement peaks decrease over time.

# Readable, exported figures

**DO**

- Export high-resolution PNG, EPS, or PDF
- Label axes and include units
- Use legends that identify each data set
- Group related cases when comparison improves the analysis

**AVOID**

- Screenshots of MATLAB or Python figure windows
- Tiny labels and unreadable legends
- Decorative plots that do not answer a question
- Separate figures when one comparison plot is clearer

**A figure should still make sense when viewed without the surrounding paragraph.**

# A figure that supports a comparison

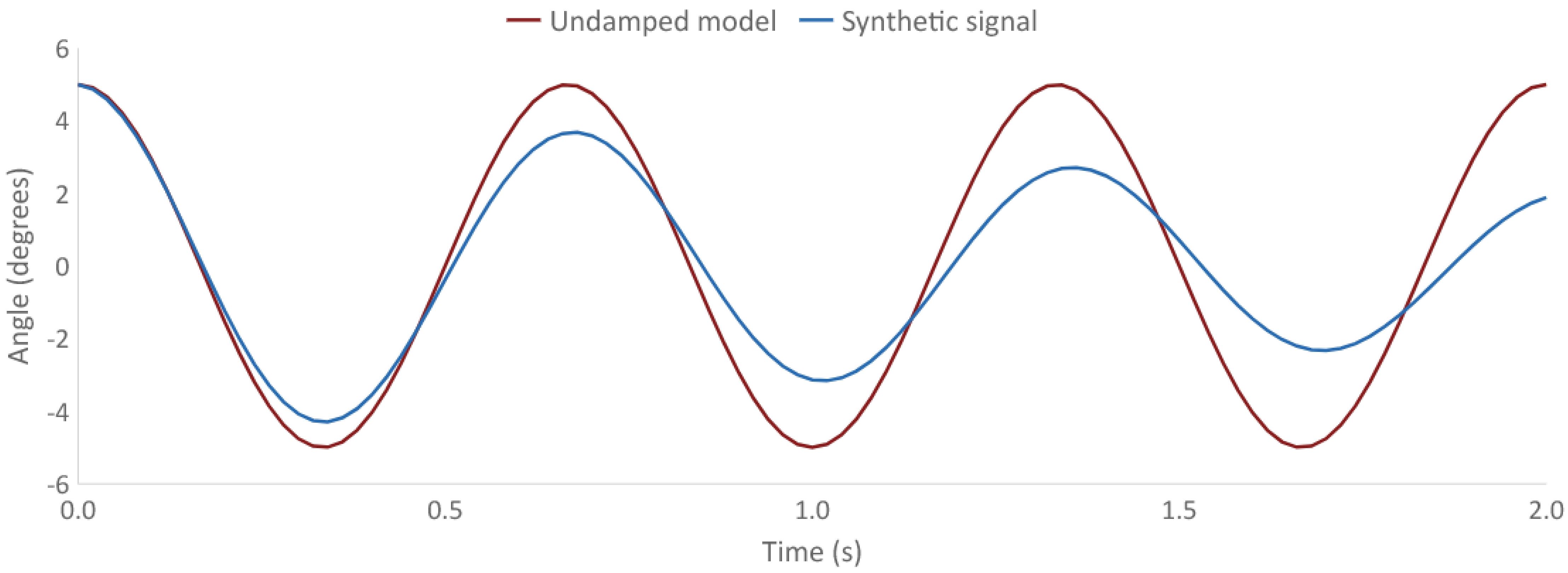



Figure 1. Illustrative angle histories: undamped model and synthetic decaying signal, 5° initial angle, 50 samples/s.

# Comparison: a fair test of the model

### Same basis

Use matching units, initial conditions, time origin, and system parameters. Explain the theoretical solution.

### Quantitative comparison

Compare period, frequency, amplitude, phase, damping, or another quantity relevant to the objective.

### Parameter honesty

State which parameters you measured, assumed, or fitted. A fit to the same data is not independent validation.

**Relative discrepancy = | measured − predicted | / | predicted | × 100%**

Use an appropriate absolute or scaled metric when the predicted value is zero or very small.

# Uncertainty and acceptable agreement

**Measurement uncertainty**

Consider calibration, resolution, repeatability, timing, and tracking choices.

**Model uncertainty**

Consider uncertain parameters and omitted physics, such as friction or large-angle effects.

**A justified conclusion**

Report what each uncertainty means. Compare the discrepancy with a consistent uncertainty estimate.

$$T = \Delta t / n_{cycles} \qquad u_T \approx u_{\Delta t} / n_{cycles}$$

Simplified timing example with exact cycle count. Longer timing intervals do not remove calibration bias.

# Discussion: explanations tied to evidence

**Observed difference**

The measured oscillation envelope decays while the undamped prediction keeps constant amplitude.

**Plausible mechanism**

Friction or drag could dissipate energy. The present comparison does not identify their separate contributions.

**Discriminating check**

Compare decay across configurations and test whether the chosen damping model describes the envelope.

Replace “human error” with a specific mechanism, its expected effect, and a way to check it.

# Conclusions, references, and appendices

## Conclusions

Answer the objective with the key numerical result, supported limitation, and a useful improvement.

## References

Cite sources where used. Include the lecture notes at minimum and list every cited source.

## Appendices

Add supporting calculations or extra data only when needed. Keep essential evidence in the main text.

# Combined reports connect related experiments

### Labs 2 and 3

Compare one angle with two coupled angles. Discuss model assumptions, initial conditions, and the limits of prediction.

### Labs 4 and 5

Relate free-response frequency and damping to the forced response. Account for configuration changes.

### Organization

Use one abstract and conclusion, with experiment-specific subsections and an explicit synthesis.

Suggested organization within the required template. Do not assume the page allowance doubles.

# The report rubric: ten equal criteria

| Each criterion is 10% of the report grade | What to check |
| --- | --- |
| Complete sections / overall aesthetic | Required organization and consistent formatting |
| Clear figures / abstract | Readable evidence and a concise, quantitative summary |
| Data analysis / introducing the theory | Traceable processing and defined governing equations |
| Introducing the experiment / comparison with theory | Physical context and a fair model-data comparison |
| Discussion / meaningful results | Evidence-based interpretation and an answer to the objective |

This table groups the ten original rubric criteria into pairs. Each individual criterion remains worth 10%.

# Submission and integrity policies

### DEADLINES

No late pre-labs.
Reports lose 25% for each day or partial day late.

### INTEGRITY

Discuss approaches, but do not copy solutions or share MATLAB, Python, or Simulink files.

### MAKE-UPS

Require instructor/GTA availability, space, and the correct equipment setup.

### VERIFICATION

Any pre-lab or report may include a brief oral check of your understanding.

**Notify the instructor during the first two weeks about approved conflicts or accommodation needs.**

# Three-minute report exercise

### Draft sentence

“The frequency was 1.47. Theory was 1.50. The error was small because of human error.”

### Work with a neighbor

Add units, calculate the discrepancy, and separate the observation from its interpretation.

### Then decide

What information is missing before you can judge agreement? Suggest one check that would help.

Invented teaching values. No submission is required for this in-class discussion.

# Exercise debrief

## Results and comparison

The measured frequency was 1.47 Hz and the predicted frequency was 1.50 Hz. Their difference was 0.03 Hz, or 2.0% of the prediction.

## Discussion

This discrepancy alone does not establish acceptable agreement. Timing and model-parameter uncertainties are needed.

## Useful next check

Verify the time calibration and repeat the frequency estimate using a stated interval and method.

Other specific, evidence-based checks can also be valid.

# The first dates

| SEP 9 | SEP 14 | SEP 21–25 | OCT 5 |
| --- | --- | --- | --- |
| **Orientation** | **Measurement fundamentals** | **Lab 1** | **Lab 1 report** |
| Course workflow and expectations | How to design and interpret measurements | Theory Monday; pre-lab due before your section | Submit on Canvas by 11:59 PM |

**There is no laboratory in Weeks 1–2.**

# Preparation for Lab 1

## BEFORE WEEK 3

- Confirm your laboratory section and GTA on Canvas
- Install or update MATLAB and Simulink
- Open the Lab 1 instructions and Pre-lab 1
- Know where and when your section meets
- Bring a way to save your experimental data

### ONE QUESTION

**What would make you trust an experimental result?**

**Questions & discussion**